

Augmentation Approaches to Refine Wearable Human Activity Recognition

Somesh Salunkhe*

University of Siegen
Siegen, Germany

somesh.salunkhe@student.uni-siegen.de

Pradnyesh Patil*

University of Siegen
Siegen, Germany

pradnyesh.patil@student.uni-siegen.de

Shubham Shinde*

University of Siegen
Siegen, Germany

shubham.shinde@student.uni-siegen.de

Robin Burchard

University of Siegen
Siegen, Germany

robin.burchard@uni-siegen.de

Abstract

In this study, we provide a detailed analysis of data augmentation techniques applied to inertial sensor data from the WEAR Dataset, aimed at improving Human Activity Recognition (HAR) in the WEAR Challenge. Augmentation techniques such as CutMix, MixUp, jittering, magnitude scaling were employed to enhance the diversity and robustness of the training dataset. These techniques aim to mitigate the challenges posed by limited and imbalanced datasets commonly encountered in wearable sensor data. We systematically applied these augmentation methods to a standard inertial dataset and evaluated their impact on the performance of various machine learning models. Our experimental results demonstrate significant improvements (about 3% increase in F1-Score and about 8% increase in Recall) in classification accuracy and model generalization, highlighting the efficacy of data augmentation in HAR applications.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing; Ubiquitous and mobile computing design and evaluation methods**; • **Hardware** → *Sensor applications and deployments*; • **Computing methodologies** → **Activity recognition and understanding; Neural networks**.

Keywords

Human Activity Recognition, Inertial Data, Data Augmentation, CutMix, MixUp, Jittering, Wearable Sensors, Machine Learning, Predictive Modeling, Classification Accuracy

*The authors marked with * contributed equally to this work

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

UbiComp Companion '24, October 5–9, 2024, Melbourne, VIC, Australia

© 2024 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 979-8-4007-1058-2/24/10

<https://doi.org/10.1145/3675094.3678464>

ACM Reference Format:

Somesh Salunkhe, Shubham Shinde, Pradnyesh Patil, and Robin Burchard. 2024. Augmentation Approaches to Refine Wearable Human Activity Recognition. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '24)*, October 5–9, 2024, Melbourne, VIC, Australia. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3675094.3678464>

1 Introduction

Recent advancements in sensor technology and the increased availability of wearable devices and mobile phones equipped with Inertial Measurement Unit (IMU) sensors have accelerated the development of Human Activity Recognition (HAR). This progress has facilitated HAR applications in various fields, including healthcare and manufacturing [3]. The easy access to sensors has resulted in an explosion of recorded data, most of which, however, remains unlabeled and thus has limited practical utility. Labeling data is challenging and involves several difficulties in recording. Additionally, advancements in GPUs have popularized the adoption of deep learning methods across almost all domains. This widespread popularity has led to the development of numerous deep learning models, making it challenging to select the best model for a given application.

Bock et al. [1] addressed these challenges by developing a well-labeled dataset named WEAR, meticulously outlining the study protocols, and evaluating the real-life performance of several popular deep learning algorithms [2]. Studies such as AutoAugHAR [6] suggest that modality-wise application of augmentation techniques can enhance performance. Another study CrossHAR [3] tackled the issue of data availability by proposing the integration of unsupervised learning with traditional deep learning algorithms. This innovative approach leverages large amounts of readily available unlabeled data for pattern recognition and then solidifies these clusters using recorded labeled data.

This study builds on the work of Bock et al., aiming to further improve the performance of the specified algorithms by incorporating data augmentation techniques as recommended by [6] and [3].

1.1 The WEAR dataset and the WEAR challenge

The “WEAR Challenge” is a competition focused on human activity recognition (HAR), organized by the creators of the WEAR

dataset[1]. Participants in this challenge are required to develop models that can accurately predict activity labels for a provided test dataset. To build and train their models, participants will utilize the WEAR dataset.

The WEAR dataset is a comprehensive collection of Inertial Measurement Unit (IMU) sensor data. It captures the movements of 18 participants as they perform 18 distinct workout routines. These routines are monitored using IMU sensors strategically placed on each of the participants' four limbs, resulting in a rich dataset of motion data.

Specifically, the dataset includes 12 features derived from IMU's accelerometer data (3-axis data from each of the four limbs). Additionally, there are 19 target classes to predict. These classes include 18 specific workout routines as well as a NULL-class that represents periods of rest or inactivity.

Participants have the freedom to choose any algorithm or machine learning approach they deem suitable for the task. The goal is to leverage the training data from the WEAR dataset to develop a predictive model that can accurately classify the activity being performed in the test dataset.

2 Data Augmentation

Data augmentation refers to a set of techniques used to artificially increase the size or improve the quality of a dataset by adding slightly modified copies of existing data or creating entirely new synthetic data. This approach is commonly used in machine learning and deep learning to improve the performance and robustness of models, especially when the original dataset is limited in size or diversity. The goal of data augmentation is to expose the model to more diverse variations of the data during training, which helps improve the model's ability to generalize to new, unseen data and reduce overfitting. By increasing the variability and quantity of data, models trained on augmented datasets often perform better on real-world data, leading to more robust and accurate predictions.

Here are some benefits of employing data augmentation in the context of HAR:

- (1) Cross-subject generalization
- (2) Availability of labeled data

2.1 Cross-subject generalization

The WEAR dataset includes data recorded from 18 participants who followed a detailed recording plan outlining the study protocols. Despite these precautions and the predetermined sensor placements, various factors will still result in differences in the recorded data. These factors include:

- (1) Variability in Exercise Execution: Since the participants were not athletes, there is intrinsic variability in how they performed the exercises, leading to different readings for the same exercise across different participants. [3]
- (2) Data Imbalance: Participants were allowed to take breaks at their discretion, determine their own break durations, and perform exercises in any order. This introduces an imbalance among the different labels in the dataset. [3]

Data augmentation helps the algorithm understand these variations in the data, enabling it to learn from them and accurately predict the correct labels across participants.

2.2 Availability of labeled data

The proliferation of low-cost wearable devices and smartphones has significantly advanced sensor-based human activity recognition (HAR), resulting in vast amounts of data. However, much of this data is unlabeled and, therefore, less useful for analysis. High-quality, well-annotated data is essential but challenging to obtain, as it requires a precise recording plan, accurate techniques to map labels to recorded sensor data, and participants who consent to have their data published. Data augmentation can help address these challenges by creating new data or generating variations in existing data, thereby enabling algorithms to learn from larger datasets.

3 Methodology

The study utilized various data augmentation techniques and sub-policies, categorizing them broadly into Label-preserving Operators and Random Transformation Operators.

Label-preserving transformation operators aim to generate synthetic data that maintain the inherent characteristics of the original data. One commonly used technique involves sample-pair-based methods, where signals and labels from two input data samples are mixed [6]. On the other hand, Random Transformation Operators introduce variability but risk altering the semantic information of the original data [6].

The study employed several augmentation techniques:

- (1) Magnitude Scaling: This technique adjusts the signal's magnitude by applying a stochastic scaling factor α , simulating variations in physical activity intensity observed in real-world scenarios. The transformed sample is given by $x' = \alpha x$, where α is a random coefficient.
- (2) Adding Noise: Also known as jittering, this method involves adding controlled amounts of random noise ϵ to data samples, thereby creating a new representation. The transformed sample is given by $x' = x + \epsilon$, where ϵ is randomly sampled noise.
- (3) MixUp: This approach generates a virtual training sample x' by linearly interpolating between two given samples x_1 and x_2 . The interpolated sample x' is computed using the formula $x' = \lambda x_1 + (1 - \lambda)x_2$, where $\lambda \in [0, 1]$ is a random coefficient.
- (4) CutMix: In CutMix, segments from two different samples x_1 and x_2 are swapped to create a new virtual sample. The resulting sample x' is a combination of parts from x_1 and x_2 defined by a binary mask M : $x' = M \odot x_1 + (1 - M) \odot x_2$, where $\odot$ denotes element-wise multiplication.

It is evident from the results of this study that no single augmentation technique is universally effective for the diverse range of sensor-based Human Activity Recognition (HAR) tasks. Building on basic augmentation techniques, this study implemented various augmentation sub-policies as suggested by Zhou et al. [6]. They noted that specific augmentation techniques perform well with certain modalities but poorly with others. Consequently, the model was extended to incorporate modality-wise data augmentation.

Given that the WEAR dataset recorded data from the limbs, it was divided into two modalities: arms and legs. Performing augmentation on a single modality or a mix of modalities is designated as a sub-policy. Following sub-policies were explored in this study:

- Data augmentation on arms modality.
- Data augmentation on legs modality.
- Data augmentation on all four limbs.

4 Experiments and Discussion

Various studies [4][5] have highlighted significant challenges in data availability and class imbalance within datasets, which can severely impact prediction scores across different labels. For example, Bock et al.'s study on the WEAR dataset observed notably low prediction scores for certain target classes, particularly the NULL-class. We hypothesize that the application of data augmentation techniques could effectively address these issues, leading to improved prediction scores across all target classes.

4.1 Dataset

The WEAR Dataset, as outlined earlier, was employed in the study. During data processing, we employed a sliding window approach with 50% overlap to segment the data. Furthermore, a separate test dataset was provided, which includes recordings of 5 new participants performing the same workout routines, as recorded by the authors.

4.2 HAR Models

In our investigation, we initially considered all four algorithms recommended by the authors of the WEAR challenge. These included two popular inertial-based algorithms and two Temporal Action Localization (TAL) models, which are typically used for video data.

(1) Inertial-based Algorithms:

- DeepConvLSTM: A deep learning model that combines convolutional neural networks (CNNs) and long short-term memory networks (LSTMs) to effectively capture spatial and temporal features from inertial data.
- Attend-and-Discriminate Model: A model that uses attention mechanisms to focus on the most relevant parts of the input data, improving classification performance.

(2) Temporal Action Localization Models (TAL):

- Tridet: A TAL model designed for detecting and localizing actions within video data.
- Actionformer: Another TAL model focused on identifying the start and end times of actions in video sequences.

We limited our study to inertial-based models (DeepConvLSTM and Attend-and-Discriminate) since the baseline performance of TAL models (ActionFormer and TriDet) was comparatively lacking. Due to their promising results, our subsequent efforts were concentrated on optimizing these models with various data augmentation strategies. This focus aimed to address the data imbalance and improve the robustness and accuracy of predictions across all target classes, including the challenging NULL-class.

5 Ablation study

The ablation study examined all possible combinations of sub-policies (data augmentation on legs, arms, and all four limbs) and data augmentation techniques (Jittering, Magnitude Scaling, CutMix, MixUp). Only two combinations yielded notable performance

gains, leading to their selection for further exploration: DeepConvLSTM with CutMix on the arm modality and Attend-and-Discriminate with MixUp on all four limbs.

5.1 DeepconvLSTM with CutMix on Arm Modality

As elaborated in section 4.2, DeepConvLSTM has gained popularity in Human Activity Recognition (HAR) applications due to its robust performance and ability to capture intricate patterns from inertial data. The model's efficacy is further enhanced through strategic data augmentation techniques. Among the various augmentation sub-policies explored, CutMix applied to the arm modality emerged as particularly effective.

In our experiments, applying CutMix to the arm modality boosted the performance of the standalone DeepconvLSTM model. This focused approach leverages the unique movement patterns captured by the arm sensors, making the augmented data more representative of real-world variations.

Table 1: Performance Comparison between Baseline and Augmented Model using DeepConvLSTM

Model	Precision	Recall	F1-Score
Baseline	80.38%	71.94%	75.42%
Augmented	77.67%	79.01%	78.21%

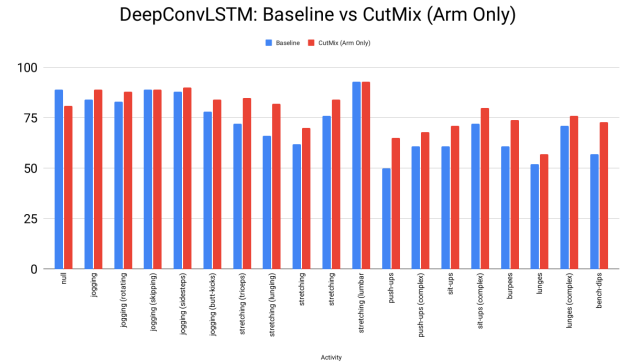


Figure 1: Comparison of label-wise F1-Scores between the baseline and CutMix augmentation applied to arm modality.

5.2 Attend-and-Discriminate with MixUp

The Attend-and-Discriminate model also showed substantial promise when combined with the MixUp augmentation strategy. The application of MixUp on all the four limbs allowed the Attend-and-Discriminate model to leverage a more diverse set of augmented samples, capturing a broader range of activity patterns. This holistic augmentation strategy proved beneficial, resulting in improved performance metrics compared to the model without augmentation. By applying MixUp on all the four limbs, the model could better

handle variations in activities, leading to more accurate and reliable predictions.

Table 2: Performance Comparison between Baseline and Augmented Model using Attend-and-Discriminate

Model	Precision	Recall	F1-Score
Baseline	78.56%	73.45%	75.66%
Augmented	78.14%	77.12%	77.51%

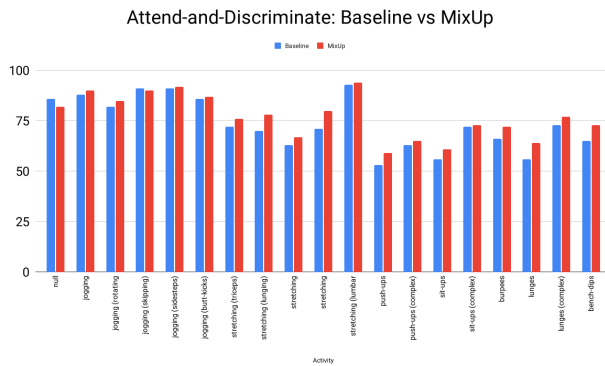


Figure 2: Comparison of label-wise F1-Scores between the baseline and MixUp augmentation applied to all four limbs.

5.3 Comparison of Results

The DeepConvLSTM model was chosen over the Attend-and-Discriminate model due to its robust performance on sequential data, and better empirical results in preliminary tests. While the Attend-and-Discriminate model showed improvements in Recall and F1-Score, the DeepConvLSTM provided more consistent and balanced improvements across all metrics.

Table 1 presents a comparative analysis of the performance between the baseline model and the augmented model using DeepConvLSTM. The metrics evaluated include Precision, Recall, and F1-Score. The augmented model demonstrates notable improvements across F1-Score and Recall compared to the baseline model.

Similarly, Table 2 compares the performance of baseline and augmented models using the Attend-and-Discriminate approach. The augmented model demonstrates a balanced improvement in Recall and F1-Score, indicating enhanced overall performance and better recognition capabilities compared to the baseline model.

These results highlight the effectiveness of the data augmentation techniques applied in enhancing the model's performance. An

improved F1-Score indicates that the model has become more proficient at distinguishing between the positive and negative classes, resulting in fewer errors of both false positives and false negatives. This improvement is crucial in scenarios where both types of misclassifications carry significant costs. Therefore, a better F1-Score reflects a model that maintains a crucial balance between Precision and Recall, leading to more reliable and effective performance in practical applications.

6 Conclusion

In this study, we aimed to improve human activity recognition using inertial sensor data through the application of data augmentation techniques. Our experiments demonstrated that the augmented DeepConvLSTM model outperformed the baseline model, achieving higher Recall and F1-Score metrics.

The choice of DeepConvLSTM was driven by its proven effectiveness in handling sequential data, superior empirical performance, computational efficiency, and scalability. The augmented model's enhanced performance underscores the importance of data augmentation in enriching training datasets and improving model robustness.

By improving the F1-Score and thereby reliability of human activity recognition models, this study contributes to some improvements in health monitoring, fitness tracking, and various other applications relying on wearable sensor technology.

Code Availability

The code used for this study is available on GitHub. You can access the repository and all related files¹.

References

- [1] Marius Bock, Hilde Kuehne, Kristof Van Laerhoven, and Michael Moeller. 2023. WEAR: An Outdoor Sports Dataset for Wearable and Egocentric Activity Recognition. arXiv:2304.05088 [cs.CV] <https://arxiv.org/abs/2304.05088>
- [2] Marius Bock, Michael Moeller, and Kristof Van Laerhoven. 2023. Temporal Action Localization for Inertial-based Human Activity Recognition. arXiv:2311.15831 [cs.LG] <https://arxiv.org/abs/2311.15831>
- [3] Zhiqing Hong, Zelong Li, Shuxin Zhong, Wenjun Lyu, Haotian Wang, Yi Ding, Tian He, and Desheng Zhang. 2024. CrossHAR: Generalizing Cross-dataset Human Activity Recognition via Hierarchical Self-Supervised Pretraining. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 8, 2, Article 64 (may 2024), 26 pages. <https://doi.org/10.1145/3659597>
- [4] Jorge-L. Reyes-Ortiz, Luca Oneto, Albert Samà, Xavier Parra, and Davide Anguita. 2016. Transition-Aware Human Activity Recognition Using Smartphones. *Neurocomputing* 171 (2016), 754–767. <https://doi.org/10.1016/j.neucom.2015.07.085>
- [5] Allan Stisen, Henrik Blunck, Sourav Bhattacharya, Thor Prentow, Mikkel Kjærgaard, Anind Dey, Tobias Sonne, and Mads Jensen. 2015. Smart Devices are Different: Assessing and Mitigating Mobile Sensing Heterogeneities for Activity Recognition. 127–140. <https://doi.org/10.1145/2809695.2809718>
- [6] Yexu Zhou, Haibin Zhao, Yiran Huang, Tobias Röddiger, Murat Kurnaz, Till Riedel, and Michael Beigl. 2024. AutoAugHAR: Automated Data Augmentation for Sensor-based Human Activity Recognition. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 8, 2, Article 48 (may 2024), 27 pages. <https://doi.org/10.1145/3659589>

¹<https://github.com/shindeshubham/wear/tree/augmentation>